

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Hariprasath
Register Number	192421246

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

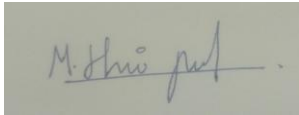
Duration : 60 Minutes

Code of Conduct

I Hariprasath(192421246) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

AIM: To Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as library book reservation system.

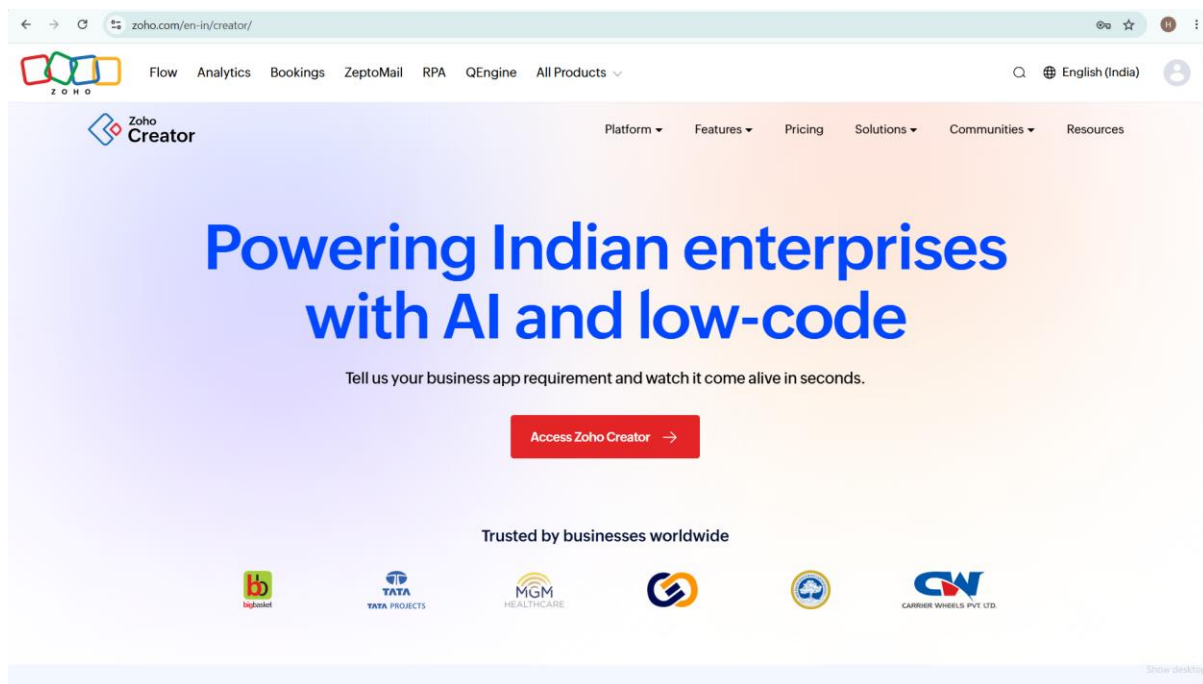
step 5: Created new application as library book reservation system.

step 6: Select one form

step 7: The software has been created.

IMPLEMENTATION:

STEP1: GOTO ZOHO.COM



STEP 2: LOGINTOTHE ZOHO.COM

STEP 3: SELECT ONE APPLICATION

Create New Application



Create using Zia
Build a functional application effortlessly using GenAI

Create



Create from scratch
Create a new application for your business needs

Create



Create from gallery
Pick a pre-built application that suits your requirement

Pick



Import from file
Import your existing data to create an application

Import

STEP 4: ENTER APPLICATION NAME

Create from scratch

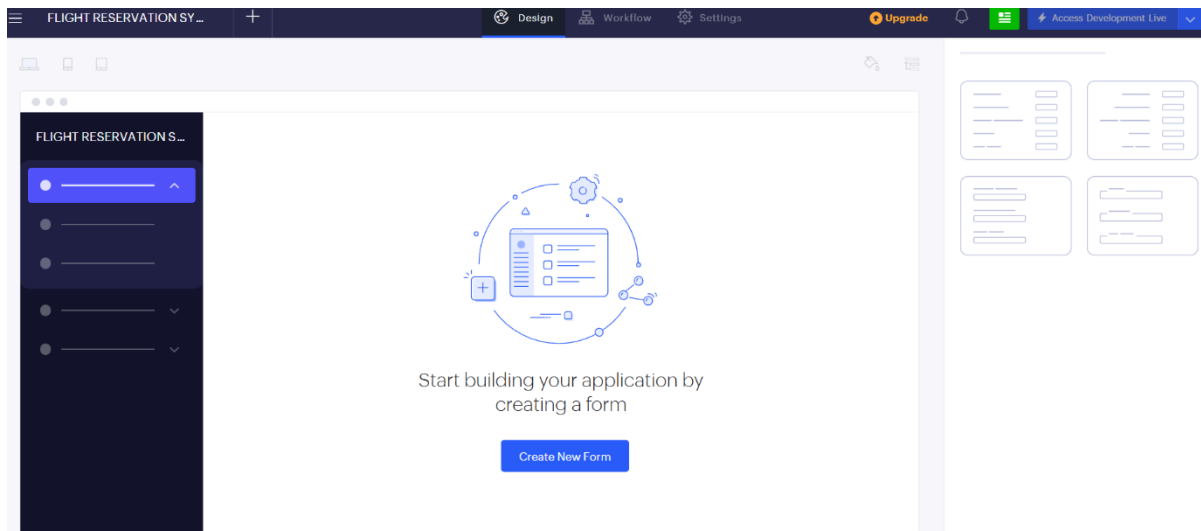
Application Name *

Enter your application name

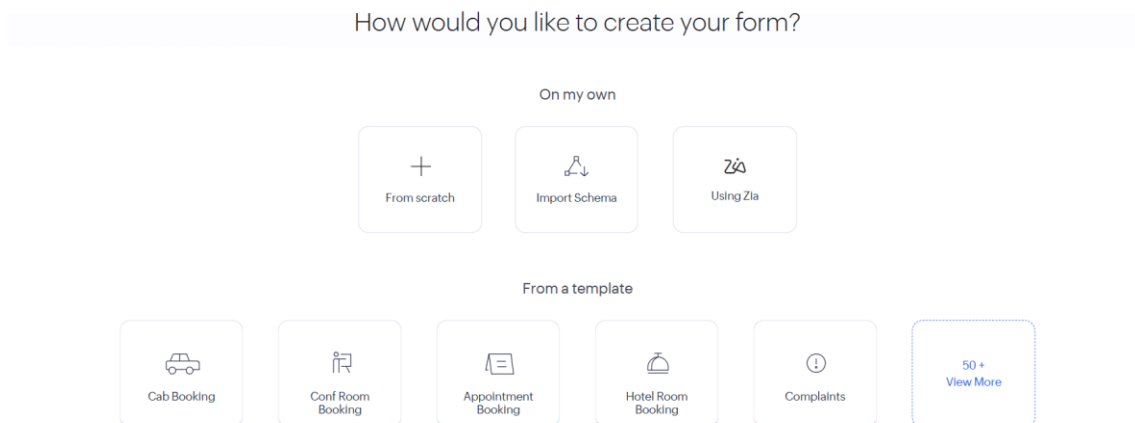
☐ Enable Environment [Learn More](#)

Create Application

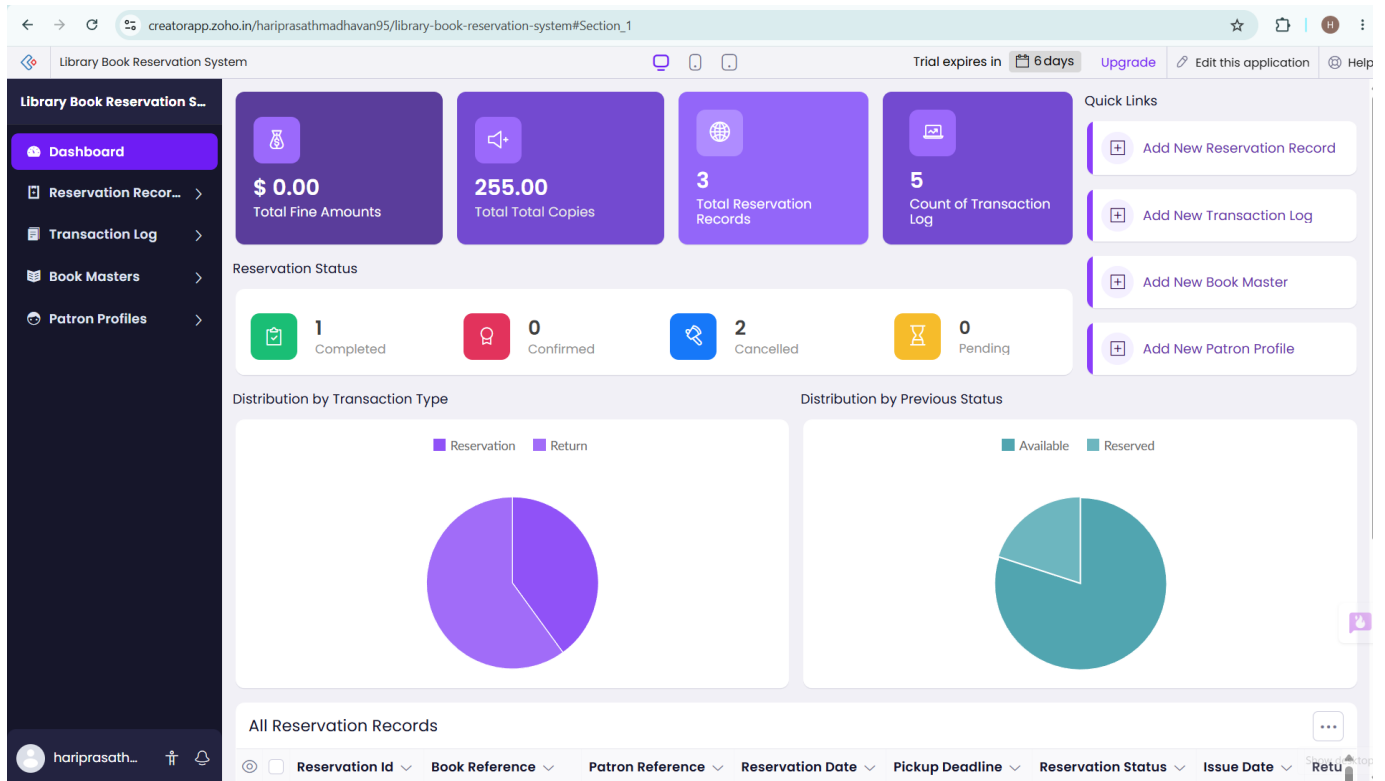
STEP 5: CREATED NEW APPLICATION



STEP 6: SELECT ONE FORM



STEP 7: THE SOFTWARE HAS BEEN CREATED.



The screenshot shows the 'Reservation Record' form in the 'Library Book Reservation System'. The form includes input fields for Reservation Id, Book Reference, Patron Reference, Reservation Date, Pickup Deadline, Reservation Status, Issue Date, Return Date, and Fine Amount. The Reservation Status field is a dropdown menu. The form also features 'Submit' and 'Reset' buttons. The left sidebar shows the 'Reservation Record' option selected. The top navigation bar includes a trial expiration notice ('Trial expires in 6 days'), an 'Upgrade' button, and links for 'Edit this application' and 'Help'.

Reservation Id	Book Reference	Patron Reference	Reservation Date	Pickup Deadline	Reservation Status	Issue Date	Return Date	Fine Amount
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Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

AIM: To Create a simple cloud software application for Property Buying & Rental process (In Chennai city) using any Cloud Service Provider to demonstrate SaaS.

PROCEDURE:

step1: Go to zoho.com.

step 2: Log into the zoho.com.

step 3: Select one application step.

step4: Enter application name as property buying & rental.

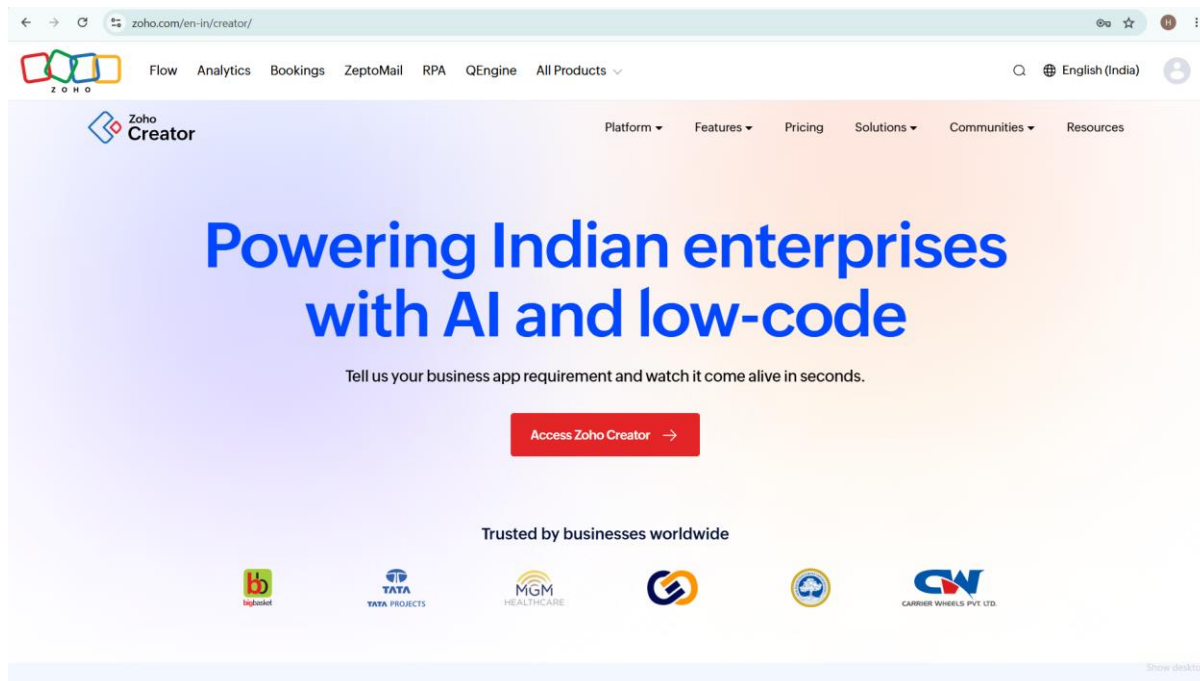
step 5: Created new application as property buying & rental.

step 6: Select one form

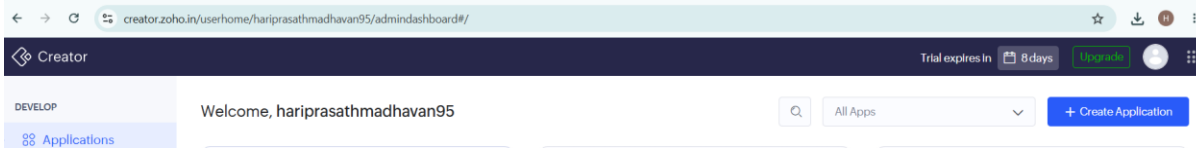
step 7: The software has been created.

IMPLEMENTATION:

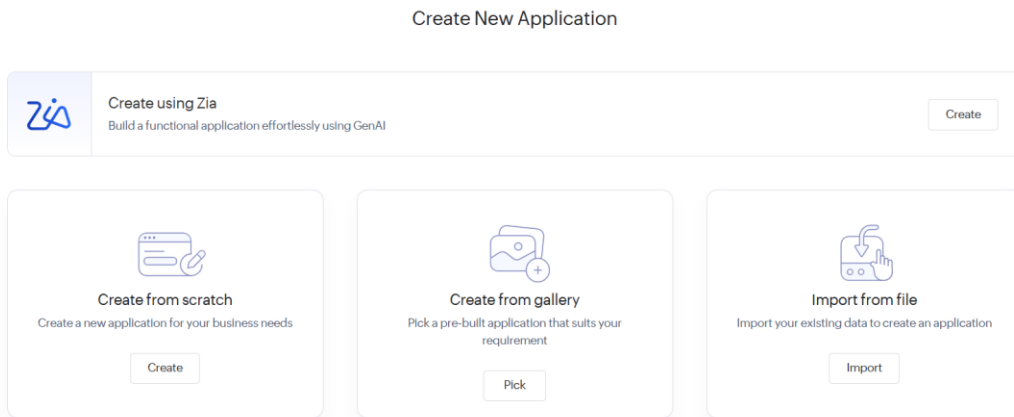
STEP1: GOTO ZOHO.COM



STEP 2: LOGINTOTHE ZOHO.COM



STEP 3: SELECT ONE APPLICATION



STEP 4: ENTER APPLICATION NAME

Create from scratch

Application Name *

☐ Enable Environment [Learn More](#)

Create Application

STEP 5: CREATED NEW APPLICATION

Payroll Processing SystemPayroll Records

DesignWorkflowSettings

Payroll Processing System

Dashboard

Employees

Payroll Configurat...

Payroll Records

+ Payroll Records

Payroll Records Re...

Board

Pay Components

192511121.sim...

Pay Components

Component Name

Component Type

Rate

Is Active

Effective From

Effective To

Submit

Reset

STEP 6: SELECT ONE FORM

How would you like to create your form?

On my own

From scratch

Import Schema

Using Zia

From a template

Cab Booking

Conf Room Booking

Appointment Booking

Hotel Room Booking

Complaints

50 +
View More

STEP 7: THE SOFTWARE HAS BEEN CREATED.

Payroll Processing System

Employees

Employee Name

First Name

Experience in Org

Overall Experience

Basic Pay

HRA

DA

TA

Gross Pay

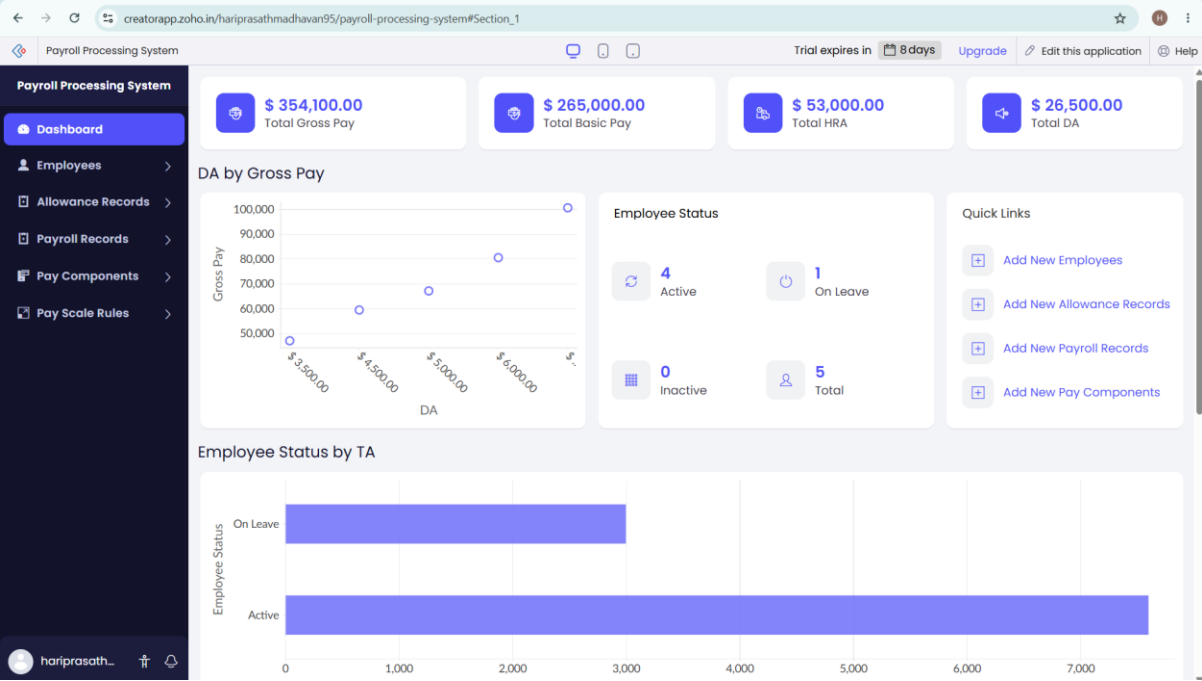
Net Pay

Pay Component Config

Employee Status

Submit

Reset



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

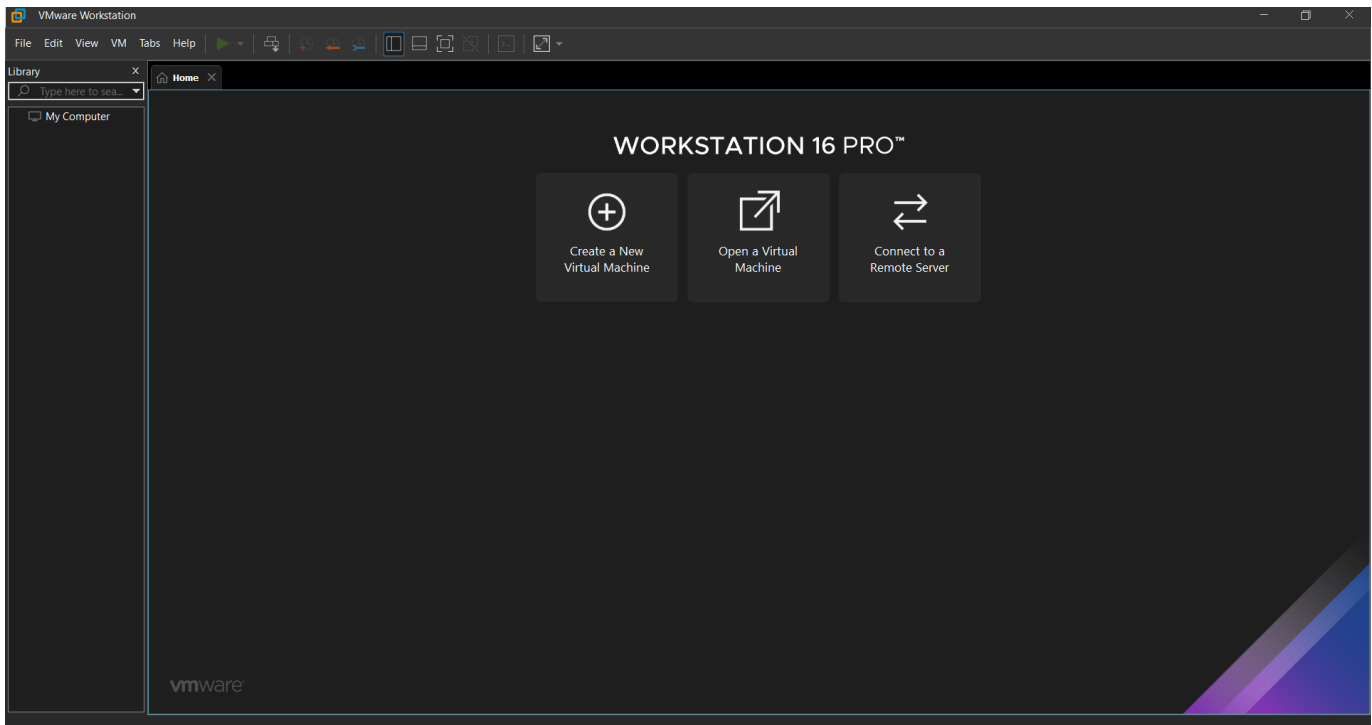
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

IMPLEMENTATION

STEP 1:DOWNLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

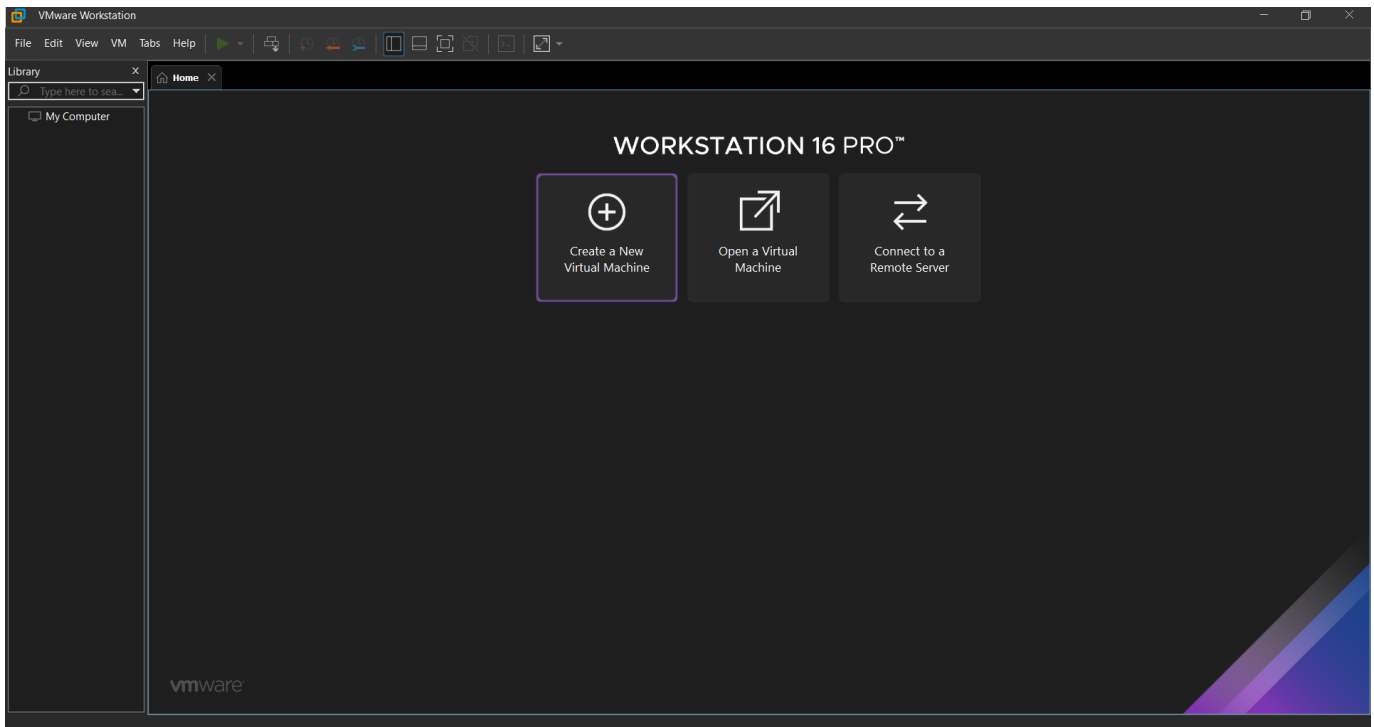


STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

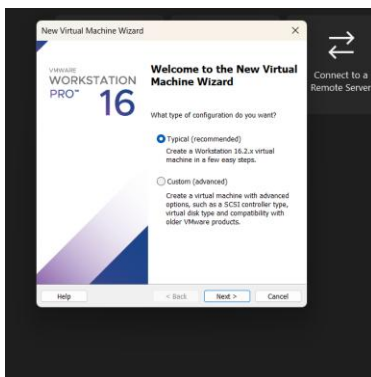
Index of /11.x/x86/release/

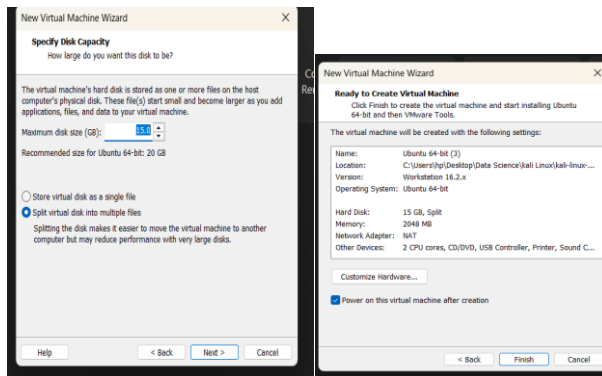
../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
Core-11.1.iso	01-Apr-2020 07:49	48
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	50639
Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	369358
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM

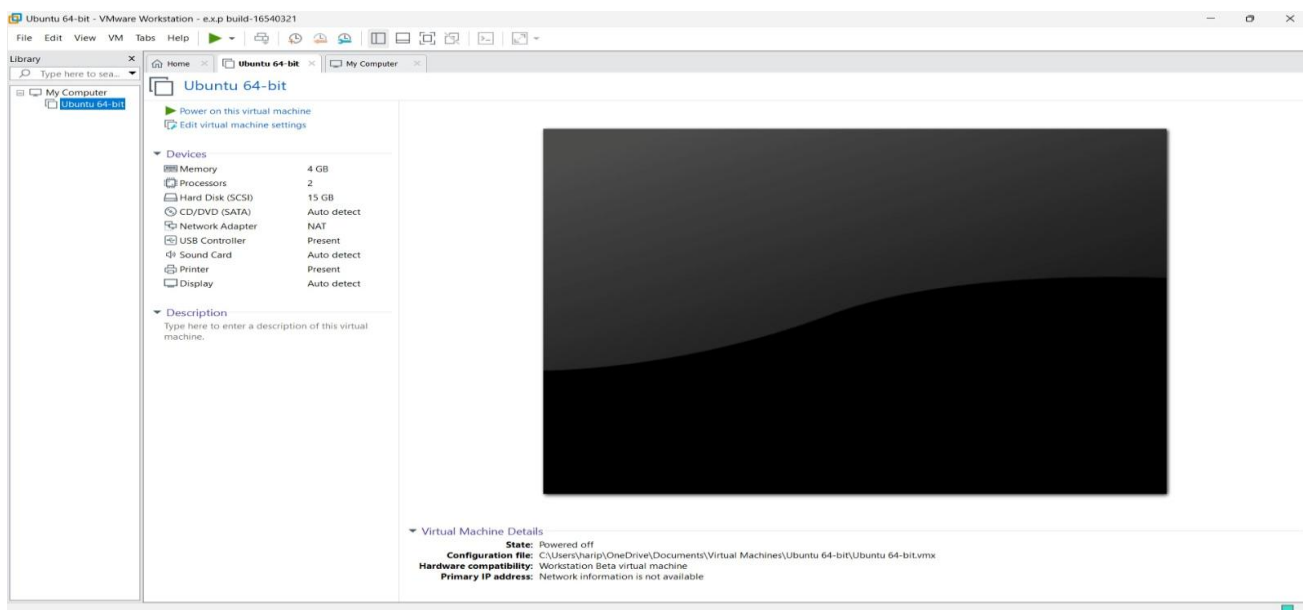


STEP 4: DO THE BASIC CONFIGURATION SETTINGS





STEP 6: Launch the VM.



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux). To create a cloning of a vm and test it by loading the previous version/cloned vm.

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

STEP 7: GO TO VM AND GOTO MANAGE AND CLICK CLONE

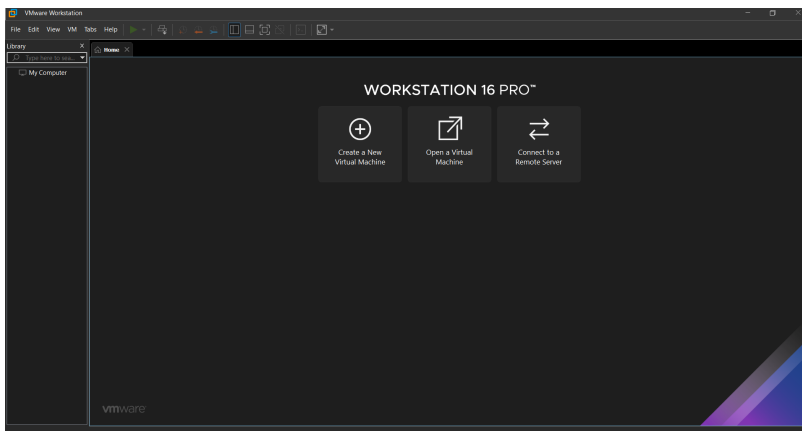
STEP 8: CLICK CLONE

STEP 9: SELECT THE FULL CLONE

STEP 10: AFTER CLONE AGAIN OR VM IS OPENED.

IMPLEMENTATION

STEP 1:DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

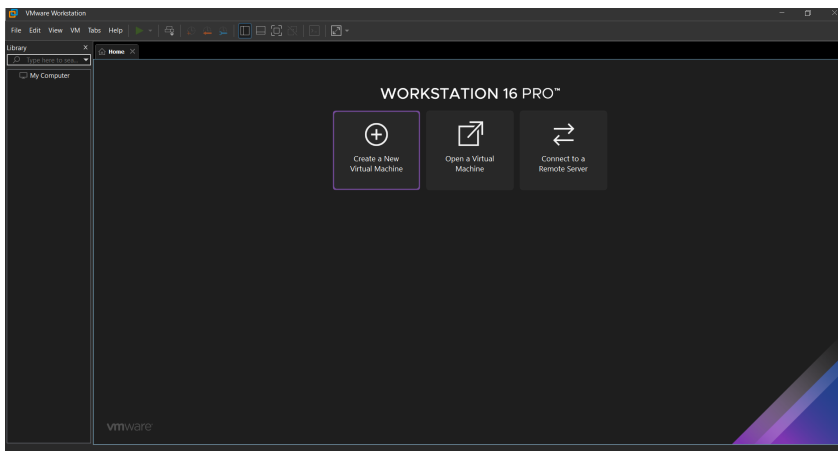


STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

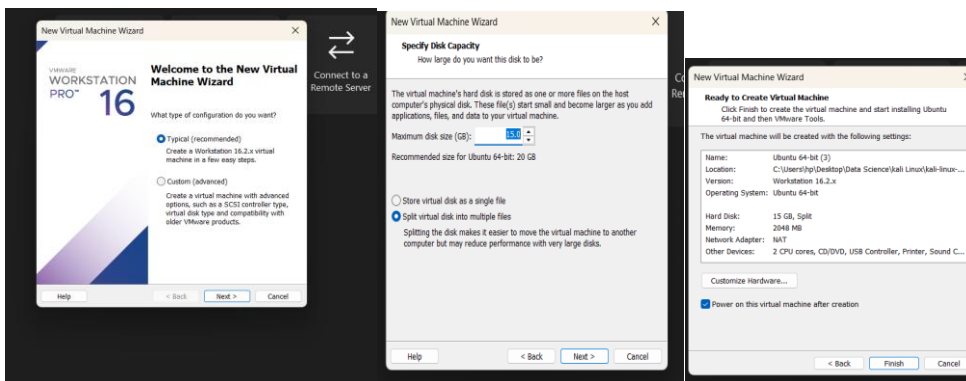
Index of /11.x/x86/release/

../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
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Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
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CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

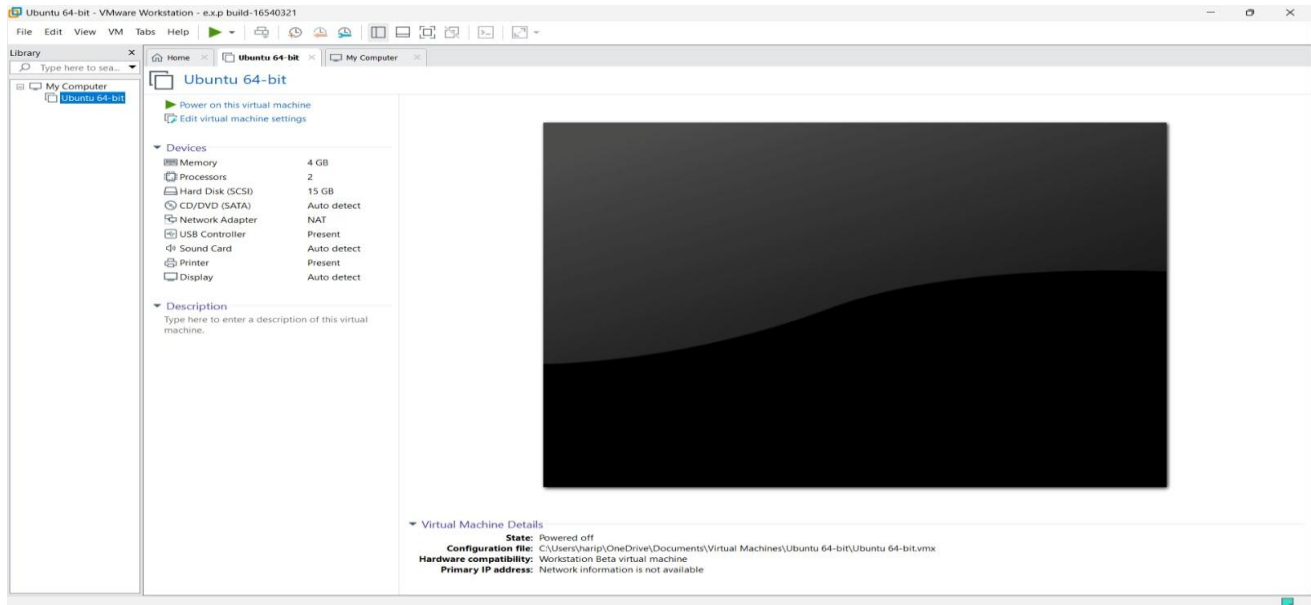
STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



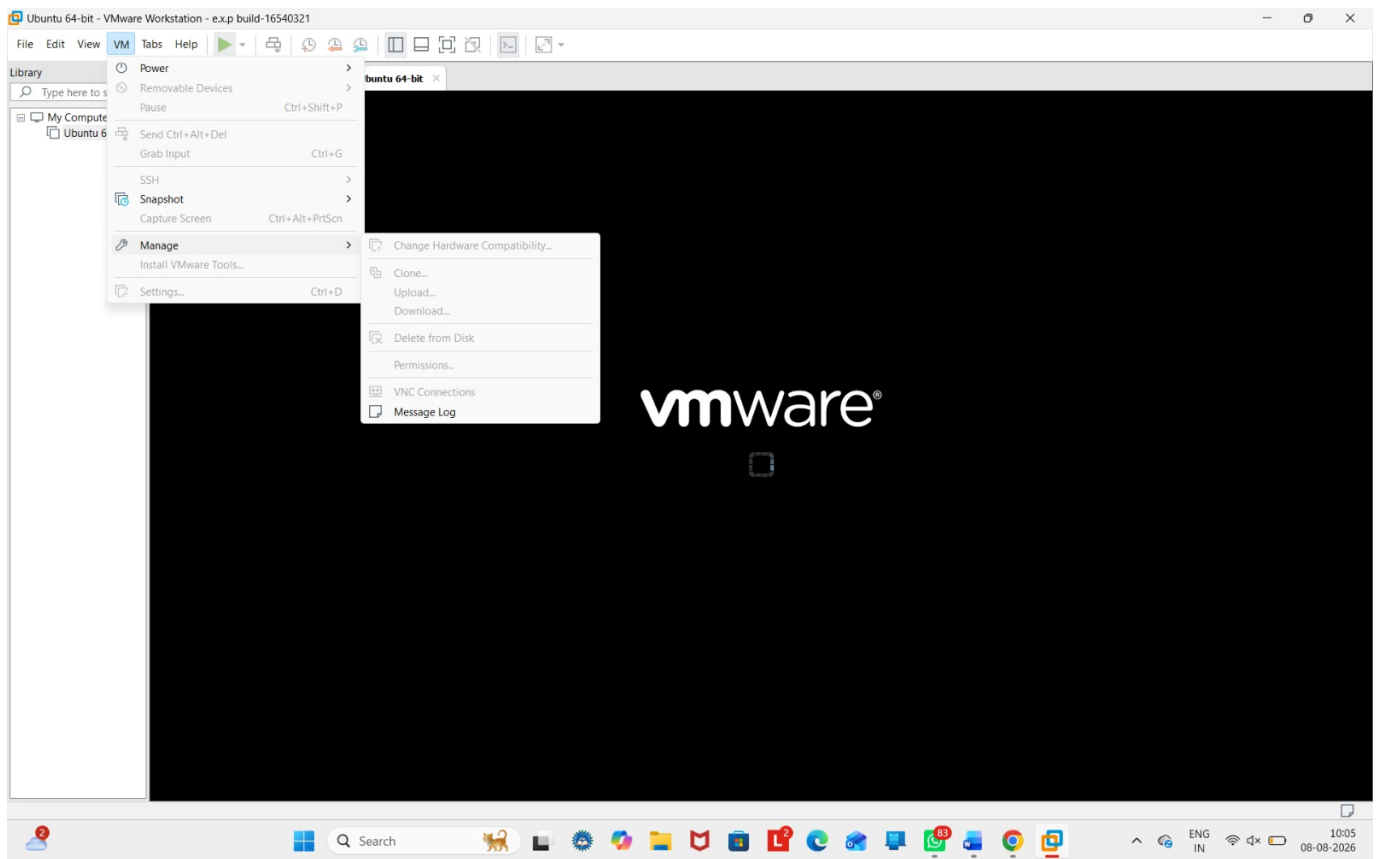
STEP 4: DO THE BASIC CONFIGURATION SETTINGS



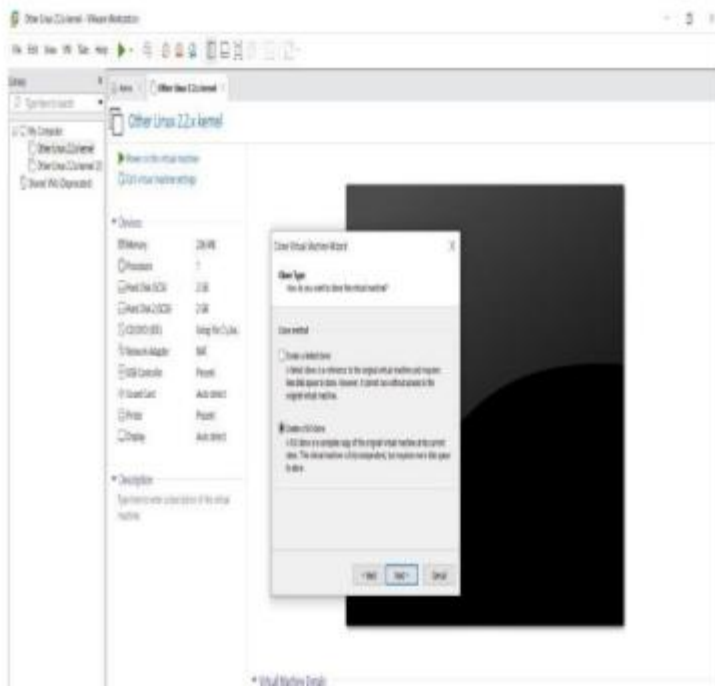
STEP 5: Launch the VM.



STEP 6: GO TO VM AND GOTO MANAGE AND CLICK CLONE



STEP 7: SELECT THE FULL CLONE



STEP 7: AFTER CLONE AGAIN OR VM IS OPENED.

